

Homework 10

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1 $2^{60} \bmod 7 = ?$

$$2^1 \bmod 7 = 2 \quad \bmod 7 = 2 \quad (1)$$

$$2^2 \bmod 7 = 4 \quad \bmod 7 = 4 \quad (2)$$

$$2^3 \bmod 7 = 8 \quad \bmod 7 = 1 \quad (3)$$

$$2^4 \bmod 7 = 16 \quad \bmod 7 = 2 \quad (4)$$

$$2^5 \bmod 7 = 32 \quad \bmod 7 = 4 \quad (5)$$

$$2^6 \bmod 7 = 64 \quad \bmod 7 = 1 \quad (6)$$

$$\dots \quad (7)$$

this pattern repeat every 3 step

$$\therefore 60 \bmod 3 = 0 \quad (8)$$

$$\therefore 2^{60} \bmod 7 = 2^3 \bmod 7 = 1 \quad (9)$$

$$\therefore 2^{60} \bmod 7 = 1$$